

Практикум E4

131 marks, 33 blocks, eight techniques. Everything in this topic is one idea used from both ends: the gradient of a curve at a point is the gradient of the straight line that touches it there. Sometimes the line is what you hand in. Sometimes the line is given and the point is what you hand in. Sometimes neither is the answer and the touching itself is the equation.

О практикуме

Вторая треть темы calculus.differentiation: 33 блока и 131 балл, всё, где производная работает наклоном. Разрез поимённый, списки лежат в blocks карточки. Ноябрьская 2023 Paper 3 посчитана корпусом дважды — зональными копиями одной бумаги из Common, — поэтому за вычетом дубля 32 блока и 126 баллов. Седьмой практикум подряд с этим дублем.

Заданий

0

Раздел решений

включён

Интерактивная версия

kaggle.com/code/artemkonukhov/ib-math-praktikum-e4-tangents

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Practicum E4: implicit differentiation, tangents and normals

131 marks, 33 blocks, eight techniques. Everything in this topic is one idea used from both ends: the gradient of a curve at a point is the gradient of the straight line that touches it there. Sometimes the line is what you hand in. Sometimes the line is given and the point is what you hand in. Sometimes neither is the answer and the touching itself is the equation.

Material. The part of `calculus.differentiation` in the AA HL archive where the derivative works as a *gradient*, sessions May 2021 — November 2025. The whole topic is 350 marks — twice what a practicum may hold — so it is cut in three by what the question asks for: *find $f'(x)$* is E3, *find the tangent* is here, *find the acceleration* is E9.

This practicum is in English, like B2 to B5 and E1 to E3. The checks speak whichever language the notebook asks them to, and this one asks for English in the setup cell.

The one thing to carry out of here.

A curve is a **condition on a pair of numbers**, not a formula for y . $y = f(x)$ is one shape that condition can take, and it is not the interesting one. Everything you know about tangents — the gradient, the normal, the point where the gradient is -1 — survives when y cannot be got out of the equation, and it survives without any change of method.

And the thing that follows from it: **the hard half of this topic is never the differentiation**. It is knowing which of the two coordinates you are short of, and going to the *curve* for it rather than to the tangent.

Where the calculator sits. 77% of the marks formally carry one, and about 16

of the 131 actually need one — two roots of $2x^2 + (\ln 2x)^2 - 2x \ln 2x - 2x = 0$, one root of $e^{2x} = 2x^2$, one root of $e^{2x}(6x - 5) = 1$, and two values of a from a cubic. This is the seventh topic in a row where that number is an illusion. Thirty of the marks are Paper 1, and that is the most honest slice this topic has ever had: a tangent to a written function is asked without a calculator, and a curve given by an equation almost always with one.

How the checks work here. All eight are new, and all eight rest on one thing: **the checks walk along the curve.**

Given a point on the curve, a check takes one step to the left and one step to the right — and for each, it solves the curve's own equation for the other coordinate. The gradient is then the gradient of the line through the two points it found. There is no $\frac{dy}{dx} = -F_x/F_y$ in there, no chain rule, no stored answer. It is the definition, walked.

check	what it holds your answer against
verify_slope	the gradient of the curve at points <i>of the curve</i>
verify_tangent	your line passes through the point and leans with it
verify_normal	the same, with the two gradients multiplying to -1
verify_where	the point is on the curve, leans right, and there are no others
verify_second	three steps along the curve instead of a second derivative
verify_right_angle	two gradients at one shared point, product -1
verify_constant	the curve is rebuilt from your letter and then walked
verify_on	the point $(a, \text{your answer})$ is on the curve

Two consequences worth knowing before you start.

Any arrangement of a line passes.

$30x - 97$, $\text{Eq}(y, 30x - 97)$ and $30(x - 4) + 23$ are one line, and the check compares lines.

Rounded answers pass, and are checked as rounded. When you write $(1.84, -0.538)$, the check does not test whether that pair is on the curve — it is not, and neither is any 3 s.f. answer. It looks for the real solution next door and asks whether your digits round to it. Get the third figure wrong and it will say so.

```

import sys
sys.path.append('..')          # from
practicum/calculus to practicum/kit.py
import sympy as sp             # the escape
hatch: anything not in kit is in sp
from kit import *              # checks +
Rational, sqrt, pi, E, log, Eq

language('en')                 # this
notebook is in English, and so are the checks

a, b, c, h, m, r = symbols('a b c h m r') #
letters that stay letters

#
x, y, t, k already come from kit

def shape(build, *answers):
    # A curve built out of your own answers.
    While they are blank there is no
    # curve to walk, and the check prints an
    empty box instead of crashing.
    return curve(...) if blank(*answers) else
    curve(build())

print('ready; sympy', sp.__version__)
print('a curve given by an equation: ',
Eq(exp(x + y), x**2 + y**2))
print('a curve that is a function:   ', Eq(y,
90*exp(-x/2)))
print('a line:                        ', 30*x
- 97)
print('a point:                       ',
(2*log(45), 2))

```

Map of techniques

#	Technique	What the question says	First move
1	Tangent at a known point	<i>find the equation of the tangent at $x = 4$</i>	get y_0 from the curve, m from f'
2	Normal	<i>normal, perpendicular to the curve</i>	$-1/m$, not $-m$
3	Gradient given, find the point	<i>gradient is $\frac{1}{3}$, parallel to $y = x$, at 70°</i>	set $f' = m$ and solve
4	The curve is an equation	$e^{x+y} = x^2 + y^2$	differentiate both sides, keep $\frac{dy}{dx}$
5	Tangent to such a curve	<i>the tangent at $x = 1$, where the tangent is horizontal</i>	you need both coordinates first
6	Differentiate the relation again	$\frac{d^2y}{dx^2}$ with no $y = f(x)$	differentiate the equation you already have
7	Right angle as a condition	<i>intersect at right angles</i>	two gradients, one point, product -1

#	Technique	What the question says	First move
8	Touching as a condition	<i>common tangent, is a tangent to, parallel to</i>	two equations, not one

The ladder goes by what the straight line is doing in the question.

Rungs 1-3 — the line is the answer.

Thirty-eight marks. The curve is a function, the work is bookkeeping, and the marks are lost on the second coordinate and on extra roots, never on the calculus.

Rungs 4-6 — the curve stopped being a function. Fifty-six marks, more than either other block. Nothing about tangents changes; what changes is that you can no longer get y by substituting x , so every gradient needs a pair.

Rungs 7-8 — the line is the condition. Thirty-seven marks. Perpendicular and tangent stop being things you find and become things you

impose, and what comes out is a constant.

The two facts the whole topic runs on.

tangent at (x_0, y_0) :

$$y - y_0 = m(x - x_0),$$

$$m = \left. \frac{dy}{dx} \right|_{(x_0, y_0)}$$

$$\begin{aligned} \text{perpendicular} &\iff m_1 m_2 \\ &= -1 \quad (\text{so the normal has } -1/m) \end{aligned}$$

Everything else in these 131 marks is getting m , and getting y_0 .

Part I — the line is the answer

Theory 1. Two numbers, and they come from different places

A tangent needs a point and a gradient. The mistake that costs whole questions is taking both from the same line of working.

$$\underbrace{y_0 = f(x_0)}$$

from the curve

$$\underbrace{m = f'(x_0)}$$

from the derivative

Three things follow, and each is worth a mark somewhere in this archive.

If a tangent is *given*, it hands you both numbers. *The line $y = 6x - 1$ is the tangent to $y = f(x)$ at $x = 4$* tells you $f'(4) = 6$ **and** $f(4) = 23$ — the second because the point of contact lies on both, so you may read $f(4)$ off the line. That is a whole mark on its own, and it is the only place in this topic where you get a value of f without knowing f .

Through a composition, the chain rule survives into the gradient. If $h = f \circ g$ then $h'(4) = f'(g(4)) \cdot g'(4)$, and $g'(4)$ is a factor people drop while concentrating on f' .

A tangent with letters in it is still just $y - y_0 = m(x - x_0)$. Paper 3 will hand you $g(x) = (x - r)(x^2 - 2ax + a^2 + b^2)$ and ask where its tangent crosses the x -axis; the algebra is unpleasant and the method is the same three lines.

Task 1 **— a tangent you are given, and a tangent you build**

November 2021 Paper 1 Q5(a)(b)(d), 5 marks · no calculator

The function f is defined for all $x \in \mathbb{R}$. The line $y = 6x - 1$ is the tangent to the graph of f at $x = 4$.

(a) Write down the value of $f'(4)$.

(b) Find $f(4)$.

The function g is defined for all $x \in \mathbb{R}$, where $g(x) = x^2 - 3x$, and $h(x) = f(g(x))$.

(c) Hence find the equation of the tangent to the graph of h at $x = 4$.

```
qla = ...          # f'(4)
qlb = ...          # f(4)
qlc = ...          # the tangent to h at x = 4,
as an expression in x

given = curve(Eq(y, 6*x - 1))          #
the tangent you were handed
verify_slope('la', qla, given, at=4)
verify_on('lb', qlb, given, 4)

# f itself is unknown, and the answer must
not depend on which f you imagine.
# So the check tries four different functions
that all satisfy f(4) = 23 and
# f'(4) = 6, composes each with g, and
demands the same tangent every time.
inner = x**2 - 3*x
family = curve(Eq(y, 6*inner - 1 + c*(inner -
4)**2))
verify_tangent('lc', qlc, family, 4, params=
{c: (0, 2, -3, 5)})
```

Task 2 — the same three lines, with letters

May 2022 TZ1 Paper 3 Q2(d)(ii), 6 marks
· Paper 3

Consider $g(x) = (x - r)(x^2 - 2ax + a^2 + b^2)$
for $x \in \mathbb{R}$, where $r, a \in \mathbb{R}$ and $b > 0$. You

may use

$$\begin{aligned} g'(x) \\ = 2(x - r)(x - a) + x^2 - 2ax + a^2 + b^2. \end{aligned}$$

(a) Find the equation of the tangent to the curve $y = g(x)$ at the point $A(a, g(a))$, as an expression in x .

(b) Hence give the x -coordinate of the point R where that tangent meets the x -axis.

The exam asks you to prove that R is $(r, 0)$. Here you find it — which is the same work, and something a check can hold.

```
q2 = ...          # the tangent at A, in terms
of x, a, b and r
q2r = ...         # the x-coordinate of R

letters = {a: (1, 2, -1, 3), b: (2, 1, 3, 1),
r: (-2, 0, 4, 1)}
cubic = curve(Eq(y, (x - r)*(x**2 - 2*a*x +
a**2 + b**2)))
verify_tangent('2a', q2, cubic, a,
params=letters)

# The second answer goes back into the first:
R is the point of your own
# tangent whose height is zero.
verify_on('2b', 0, shape(lambda: Eq(y, q2),
q2), q2r, params=letters)
```

Theory 2. The normal, and the one step where it differs

A normal is perpendicular to the tangent at the same point, so

$$m_{\text{normal}} = -\frac{1}{m_{\text{tangent}}}.$$

Not $-m$. The archive contains that slip in both directions, and it is invisible in your own working because -2 and $-\frac{1}{2}$ look equally plausible sitting next to a gradient of 2.

Two corners worth naming.

Where the tangent is horizontal the normal is vertical, and a vertical line has no gradient at all — its equation is $x = c$, and $y = mx + c$ cannot express it.

A normal with a letter in it is where this topic starts being interesting.

For $y = \frac{1}{x}$ the tangent at $x = t$ has gradient $-\frac{1}{t^2}$, so the normal has gradient t^2 — and now the *whole family* of normals is parametrised by t , which is what

November 2025 Paper 3 spends 26 marks on.

Task 3 — a normal, printed in the question

November 2025 TZ3 Paper 1 Q9(c), 5 marks · no calculator

Consider $f(x) = \frac{1}{2}x^2 + kx + 13$ with $k = 5$.
The line L is normal to the curve $y = f(x)$ at $x = -3$.

Show that the equation of L is $y = -\frac{1}{2}x + 1$ by writing it down as an expression in x .

```
q3 = ...          # the normal at x = -3, as  
an expression in x
```

```
verify_normal('3', q3, curve(Eq(y, x**2/2 +  
5*x + 13)), -3)
```

Task 4 — a normal with a letter for the point

November 2025 TZ1 Paper 3 Q1(a)(i), 2 marks · Paper 3

The curve H has equation $y = \frac{1}{x}$, where $x \in \mathbb{R}$, $x \neq 0$. A line N is normal to H at $x = t$.

Show that the gradient of N is t^2 by writing that gradient down.

```
q4 = ...           # the gradient of N, in
terms of t

# You hand in a gradient; the check builds
the normal through the point of H
# out of it and holds that line against the
curve.
line = ... if blank(q4) else q4*(x - t) + 1/t
verify_normal('4', line, curve(Eq(y, 1/x)),
t, params={t: (0.5, 1, 2, 3)})
```

Theory 3. Backwards: the gradient is given, the point is not

Three phrasings, one meaning. Convert first, solve second.

the question says	the gradient is
<i>the gradient of L is $\frac{1}{3}$</i>	$\frac{1}{3}$
<i>the tangent is parallel to $y = x$</i>	1
<i>the tangent makes an angle of 70° with the horizontal</i>	$\tan 70^\circ$

Then solve $f'(x) = m$, and after that three habits:

Throw out the roots the domain forbids. $f(x) = \ln(x^2 - 16)$ is given for $x > 4$; $f'(x) = \frac{1}{3}$ has roots 8 and -2 , and the mark scheme says in so many words that including -2 loses the mark. Extra answers are not free.

If the question says *coordinates*, give two numbers. Substituting back into f is a separate mark, every time.

If the question says *exact*, do not touch the calculator at the end. $2 \ln 45$ is exact; 7.61 is not, and it scores zero on a question that asked for exact coordinates even though it is the same number.

Task 5 — one root wanted, two available

May 2021 TZ2 Paper 1 Q4(b), 6 marks · no calculator

Consider $f(x) = \ln(x^2 - 16)$ for $x > 4$. The line L is the tangent to the graph of f at the point B, and the gradient of L is $\frac{1}{3}$.

Find the x -coordinate of B.

```
q5 = ...          # the x-coordinate of B – a
                  # list, if you think there is more than one

# coordinates=False: this question asked for
# x alone, so y is not held against
# you. The domain from the question is passed
# in, and any answer outside it
# comes back as an extra root, which is
# exactly what the mark scheme does.
verify_where('5', q5, curve(Eq(y, log(x**2 -
16))), Rational(1, 3), (4.01, 30),
              coordinates=False)
```

Task 6 — exact coordinates

May 2021 TZ1 Paper 2 Q10(b), 4 marks · calculator allowed

Consider $f(x) = 90e^{-0.5x}$ for $x \in \mathbb{R}^+$. The line L has gradient -1 and is tangent to

the graph of f at the point Q.

Find the **exact** coordinates of Q.

```
q6 = ...          # (x, y) – exact, both of  
them
```

```
verify_where('6', q6, curve(Eq(y, 90*exp(-  
x/2))), -1, (0.01, 20))
```

Task 7 — the gradient arrives as an angle

May 2025 TZ3 Paper 2 Q5(b), 4 marks · calculator

Consider $f(x) = \frac{(2x + a)^3}{(x + 5)^2}$, where $x \neq -5$ and $a \in \mathbb{R}^+$. When $x = 1$, the tangent to the graph of f makes an angle of 70° with the horizontal.

Find the two possible values of a , to 3 significant figures.

```
q7 = ...          # a list of both values

# The letter sits inside the curve, so there
# is nothing to substitute the
# answer into. The check builds the curve out
# of each value you give and then
# walks it – and sweeps the window for values
# you missed.
verify_constant('7', q7, lambda v:
curve(Eq(y, (2*x + v)**3/(x + 5)**2)),
      1, tan(70*pi/180), (0.1, 25))
```

Part II — the curve is an equation

Theory 4. y is a function of x even when nobody wrote it that way

$e^{x+y} = x^2 + y^2$ cannot be solved for y . It still describes a curve, and that curve still has a gradient at each of its points. The only new rule is the chain rule, applied to y :

$$\begin{aligned}\frac{d}{dx}(y^2) &= 2y \frac{dy}{dx}, \\ \frac{d}{dx}(xy) &= y + x \frac{dy}{dx}, \\ \frac{d}{dx}(e^{x+y}) &= e^{x+y} \left(1 + \frac{dy}{dx}\right)\end{aligned}$$

Then the same four moves every time: differentiate both sides, collect the $\frac{dy}{dx}$ terms on one side, factor, divide.

The answer has both letters in it, and that is not a failure. $\frac{4(3-x)}{y+2}$ is a complete answer; there is nothing further to do with it. It also means that to get a

number you need a *point*, both coordinates — which is the whole difficulty of the next rung.

When x is in the base and the exponent at once, take logs first.

$g(x) = (S/x)^x$ has no rule that applies. Writing $\ln g = x \ln(S/x)$ and differentiating both sides gives $\frac{g'}{g} = \ln S - \ln x - 1$, and since $g \neq 0$, setting $g' = 0$ is setting that bracket to zero. This is implicit differentiation used on a function that was explicit — the technique is about what is convenient, not about what is possible.

Task 8 — an ellipse, and the printed line

May 2025 TZ2 Paper 2 Q12(a), 4 marks · calculator

The curve C has equation $4x^2 + y^2 - 24x + 4y + 20 = 0$.

Use implicit differentiation to show that $\frac{dy}{dx} = \frac{4(3-x)}{y+2}$, by writing the gradient

down as an expression in x and y .

```
q8 = ...           # dy/dx, in terms of x and y

# No point is named, so the check picks
# several points of the curve itself and
# demands that one written expression give
# the right gradient at all of them.
verify_slope('8', q8, curve(4*x**2 + y**2 -
24*x + 4*y + 20), domain=(1.5, 4.5))
```

Task 9 — a logarithm of a product, then a tangent

November 2021 Paper 2 Q8, 8 marks · calculator

Consider the curve C given by $y = x - xy \ln(xy)$, where $x > 0$, $y > 0$.

(a) The exam asks you to show that

$$\frac{dy}{dx} + \left(x \frac{dy}{dx} + y \right) (1 + \ln(xy)) = 1.$$

Do that, and then finish the job: write $\frac{dy}{dx}$ as an expression in x and y .

(b) Hence find the equation of the tangent to C at the point where $x = 1$, as an expression in x .

```

q9a = ...          # dy/dx, in terms of x and y
q9b = ...          # the tangent to C at x = 1,
as an expression in x

spiral = curve(Eq(y, x - x*y*log(x*y)))
verify_slope('9a', q9a, spiral, domain=(0.4,
1.6))
verify_tangent('9b', q9b, spiral, (1, 1))

```

Theory 5. To get a number out of $\frac{dy}{dx}$ you need a pair

This is the whole of rung 5, and it cuts both ways.

Forwards. *The tangent at the point where $x = 1$.* You have one coordinate; the other comes from the **curve's own equation** with $x = 1$ substituted — here $y = 1 - y \ln y$, which has the single solution $y = 1$. Only then does $\frac{dy}{dx}$ become a number.

Backwards. *Where is the tangent horizontal?* Setting a fraction to zero means setting its **numerator** to zero, and that is one equation in two unknowns. The second equation is the curve. Solve the pair.

$$2x - e^{x+y} = 0 \quad \text{and} \quad e^{x+y} = x^2 + y^2$$

$$\implies y = \ln(2x) - x$$

$$\implies 2x^2 + (\ln 2x)^2 - 2x \ln 2x - 2x = 0,$$

and *that* is what the calculator is for. Two roots, so two points — and stopping at one is the commonest lost mark of the rung.

A gradient of -1 on a curve symmetric in $y = x$ is a special case worth seeing. $\frac{dy}{dx} = -1$ forces $y = x$ here, and substituting that into the curve leaves one equation in one unknown.

Task 10 — the same curve, three ways

May 2024 TZ1 Paper 2 Q11(a)(b)(d), 18 marks · calculator

The curve C is defined by $e^{x+y} = x^2 + y^2$ and has a line of symmetry $y = x$. There are two points on C where the tangent is horizontal, labelled P and Q.

(a) Show that $\frac{dy}{dx} = \frac{2x - e^{x+y}}{e^{x+y} - 2y}$ by writing it down.

(b) Find the coordinates of P and of Q, to 3 significant figures.

(c) Find the coordinates of the point on C where the tangent has gradient -1 , to 3 significant figures.

```
q10a = ...      # dy/dx, in terms of x and y
q10b = ...      # [(x, y), (x, y)] – both
points, 3 s.f.
q10d = ...      # (x, y) – the point where
the gradient is -1, 3 s.f.

loop = curve(Eq(exp(x + y), x**2 + y**2))
verify_slope('l0a', q10a, loop, domain=(0.2,
2))
verify_where('l0b', q10b, loop, 0, (0.05, 3))
verify_where('l0c', q10d, loop, -1, (-1,
0.5))
```

Theory 6. Differentiating the relation a second time

When there is no $y = f(x)$, there is nothing to differentiate twice — except the equation you already have.

Differentiate it again, treating both y and $\frac{dy}{dx}$ as functions of x :

$$\begin{aligned}\frac{d}{dx} \left(x \frac{dy}{dx} \right) &= \frac{dy}{dx} + x \frac{d^2y}{dx^2}, & \frac{d}{dx} \left(y \frac{dy}{dx} \right) \\ &= \left(\frac{dy}{dx} \right)^2 + y \frac{d^2y}{dx^2}\end{aligned}$$

The second of these is where $\left(\frac{dy}{dx}\right)^2$ comes from, and it is the term people lose.

Substitute numbers last. Differentiate the whole relation, *then* put in $x = 1$, $y = \frac{3}{2}$, $\frac{dy}{dx} = -1.7$. Substituting first destroys the variable you were about to differentiate.

The same move solves a differential equation you have never solved. $\frac{dP}{dt} = kP(1 - P/N)$ gives $\frac{d^2P}{dt^2}$ by the product rule plus one substitution of the first equation back into itself — and that second derivative is what tells you the population grows fastest at $P = N/2$.

Task 11 — a second derivative with no formula for y

November 2025 TZ3 Paper 2 Q12(a)(i)(b)(ii), 6 marks · calculator

Consider the homogeneous differential equation

$$\begin{aligned}(x^2 + xy) \frac{dy}{dx} &= x^2 \\ &+ xy \\ &- 3y^2, \\ x &> 0, \\ y &> \frac{x}{2},\end{aligned}$$

with $y = \frac{3}{2}$ when $x = 1$.

(a) Find the value of $\frac{dy}{dx}$ when $x = 1$.

(b) Differentiating the relation again gives

$$\begin{aligned}(x^2 + xy) \frac{d^2y}{dx^2} &= 2x \\ &+ y \\ &- x \left(\frac{dy}{dx} \right)^2 \\ &- (x + 7y) \frac{dy}{dx}.\end{aligned}$$

Find the value of $\frac{d^2y}{dx^2}$ when $x = 1$, exactly.

*Part (d) of that question solves the equation and gets $x^6(2y - x)^3 = 2(x + 2y)$. The checks below walk **that** curve — the one your solution is a point of — so nothing they do repeats the work you are being asked to do.*

```
q11a = ...          # dy/dx at x = 1
q11b = ...          # the second derivative at x
                    = 1, exactly

solved = curve(Eq(x**6*(2*y - x)**3, 2*(x +
2*y)))
verify_slope('11a', q11a, solved, at=(1,
Rational(3, 2)))
verify_second('11b', q11b, solved, (1,
Rational(3, 2)))
```

Part III — the line is the condition

Theory 7. A right angle is an equation

Show that the curves intersect at right angles is not a request for a line. It is three separate claims, and the mark scheme pays for all three:

1. the point lies on **both** curves;
2. each gradient is the gradient of **its own** curve there;
3. the two multiply to -1 .

The failure is almost never the differentiation. It is substituting the point into one curve's gradient and a different point into the other's, or stopping after showing the two gradients are different.

Families are where this gets used.

Each member of the family $y = mx$ meets each member of family C at right angles.

The gradient of $y = mx$ at the point (x, y) on it is $m = \frac{y}{x}$ — written in x and y , not in

m , which is a mark on its own and the reason the whole thing works. Then perpendicularity says the other family has $\frac{dy}{dx} = -\frac{x}{y}$, and solving *that* differential equation gives $x^2 + y^2 = k$: circles, cutting every line through the origin square.

Task 12 — two curves, one point, one right angle

*May 2023 TZ2 Paper 1 Q8(a)(b), 4 marks
· no calculator*

The functions f and g are defined by $f(x) = \cos x$ for $0 \leq x \leq \frac{\pi}{2}$ and $g(x) = \tan x$ for $0 \leq x < \frac{\pi}{2}$. The curves meet at a point P whose x -coordinate is k .

Given that $\cos^2 k = \sin k$, show that the tangent to $y = f(x)$ at P and the tangent to $y = g(x)$ at P meet at right angles — by writing down the two gradients in terms of k .

```

q12f = ...      # f'(k)
q12g = ...      # g'(k)

meet = nsolve(cos(x) - tan(x), x, 0.6)  #
where the two curves cross
verify_right_angle('12', (q12f, q12g),
curve(Eq(y, cos(x))), curve(Eq(y, tan(x))),
(k, cos(k)), params={k:
(meet,)}))

```

Task 13 — two families, cutting square everywhere

November 2023 Paper 3 Q2(a)(i)(d), 6 marks · Paper 3

(a) Consider the family of straight lines L with equation $y = mx$, where m is a parameter. Write down an expression for the gradient of L in terms of x and y .

A family of curves has equation $y^2 = 4a^2 - 4ax$, where $a > 0$; a second family has $y^2 = 4b^2 + 4bx$, where $b > 0$. They meet at $M(a - b, 2\sqrt{ab})$.

(b) Show that at M the two curves intersect at right angles, by writing down the two gradients in terms of a and b .

```

q13a = ...          # the gradient of L, in
terms of x and y
q13f = ...          # dy/dx of the first family
at M
q13g = ...          # dy/dx of the second family
at M

verify_slope('l3a', q13a, curve(Eq(y, m*x)),
domain=(0.5, 3), params={m: (2, -1, 0.5)})

first = curve(Eq(y**2, 4*a**2 - 4*a*x))
second = curve(Eq(y**2, 4*b**2 + 4*b*x))
verify_right_angle('l3b', (q13f, q13g),
first, second, (a - b, 2*sqrt(a*b)),
                        params={a: (2, 3, 5), b:
(1, 2, 4)})

```

Theory 8. Touching is two equations, not one

The graphs of f and g have a common tangent at $x = 3$. That sentence is worth two equations:

$$f'(3) = g'(3) \quad \text{and} \quad f(3) = g(3).$$

The gradients alone give parallel, not touching. The values alone give crossing, not touching. Both together give touching, and questions of this kind hand

you exactly as many unknowns as you have equations.

Two other faces of the same idea:

Parallel to a given line is only the first equation, and that is enough when the question only wants one constant: *the tangent to $y = (2x - 1)e^{kx}$ at $x = 1$ is parallel to $y = 5e^kx$ gives $e^k(k + 2) = 5e^k$, and since $e^k \neq 0$ — say so — $k = 3$.*

A double root is touching. If $f(x) - \ell(x)$ factors as $(x - 4)^2(x - 1)$ then the line ℓ meets the curve twice at $x = 4$, which is what tangency means. It is an alternative proof, it is on the mark scheme, and on a cubic it is faster than differentiating.

And exact still means exact. $a = e^{1/e}$ is the answer; 1.44 is the check that you have not blundered.

Task 14 — a common tangent fixes two constants

*May 2021 TZ1 Paper 1 Q4(b)(c), 6 marks
· no calculator*

Consider $f(x) = -(x - h)^2 + 2k$ and $g(x) = e^{x-2} + k$, where $h, k \in \mathbb{R}$. The graphs of f and g have a common tangent at $x = 3$.

(a) Show that $h = \frac{e+6}{2}$, by writing h down.

(b) Hence show that $k = e + \frac{e^2}{4}$, by writing k down.

Note. The corpus this practicum is built from prints $k = \frac{2e+e^2}{4} \approx 3.21$ here. The paper prints $e + \frac{e^2}{4} \approx 4.57$, and the paper is right.

```

q14h = ...      # h
q14k = ...      # k

# (a) is the first of the two conditions: the
# gradient of f at x = 3 equals
# g'(3) = e. The check rebuilds f from your h
# and walks it.
verify_constant('l4a', q14h, lambda v:
curve(Eq(y, -(x - v)**2)), 3, E, (0, 8))

# (b) is the second condition, and both
# answers go into it: the point that g
# passes through at x = 3 has to lie on f as
# well.
verify_on('l4b', ... if blank(q14k) else E +
q14k,
          shape(lambda: Eq(y, -(x - q14h)**2
+ 2*q14k), q14h, q14k), 3)

```

Task 15 — the line is given, the curve is the unknown

May 2023 TZ2 Paper 3 Q1(e), 8 marks · Paper 3

For $1.4 \leq a \leq 1.5$ there is a value of a for which the line $y = x$ is a tangent to the graph of $y = \log_a x$ at a point P.

Find the **exact** coordinates of P and the **exact** value of a .

```
q15x = ...      # the x-coordinate of the
tangency point, exactly
q15y = ...      # the y-coordinate of the
tangency point, exactly
q15a = ...      # a, exactly

# Everything the question gave is what comes
back out: the check builds the
# curve from your a, takes your own point of
contact, and asks whether the
# line  $y = x$  – the one line you were given –
is tangent to it there.
verify_tangent('15', x, shape(lambda: Eq(y,
log(x)/log(q15a)), q15a),
               (q15x, q15y))
```

Trainer: name the technique in five seconds

Twelve questions from the archive, none of them solved. Say which technique each one is, in one word, **before** doing any of them.

code	technique
tangent	a tangent at a point the question names
normal	the word <i>normal</i> , or perpendicular to the curve
point	the gradient is given; the point is the answer
implicit	the curve is an equation; the answer keeps both letters
itangent t	a tangent or a point on such a curve
second	differentiate the relation a second time
right	two curves, one point, product -1
conditi on	touching is the claim, and a constant comes out

1. Show that $\frac{dy}{dx} = \pm \frac{3x^2 + 1}{2\sqrt{x^3 + x}}$ for $x > 0$ on $y^2 = x^3 + x$.
2. The tangent to $y = (2x - 1)e^{kx}$ at $x = 1$ is parallel to $y = 5e^k x$. Find k .
3. Find the coordinates of the point on $y = e^{2x}(3x - 4)$ where the tangent is parallel to $y = x$.
4. Show that the gradient of the line normal to $y = \frac{1}{x}$ at $x = t$ is t^2 .

5. Find the gradient of the tangent T_k to $y = 15 \cos\left(\frac{\pi x}{50}\right) + 15$ at $(k, h(k))$, in terms of k .
6. Show that $\frac{d^2 P}{dt^2} = k^2 P \left(1 - \frac{P}{N}\right) \left(1 - \frac{2P}{N}\right)$, given $\frac{dP}{dt} = kP \left(1 - \frac{P}{N}\right)$.
7. At M, show that $y^2 = 4a^2 - 4ax$ and $y^2 = 4b^2 + 4bx$ intersect at right angles.
8. Find the equation of the tangent to $y^2 = x^3 + 2$ at the rational point $P(-1, -1)$.
9. Given that the tangent to $y = \ln(x^2 - 16)$ at B has gradient $\frac{1}{3}$, find the x -coordinate of B.
0. Show that the line $y = x - 1$ is tangent to $y = (x - 1)(x^2 - 8x + 17)$ at $A(4, 3)$.
1. The line $y = 6x - 1$ is the tangent to $y = f(x)$ at $x = 4$, and $h(x) = f(x^2 - 3x)$. Find the tangent to h at $x = 4$.
2. Find, in terms of S , the x -coordinate of the maximum of g , where $\ln(g(x)) = x \ln\left(\frac{S}{x}\right)$.

```
answers = {
    1: '',    2: '',    3: '',    4: '',
    5: '',    6: '',    7: '',    8: '',
    9: '',   10: '',   11: '',   12: '',
}

KEY = {1: '3a8f7b6c00a3', 2: '596d40a4ed24',
3: '251fec592e5', 4: '317b32c14369', 5:
'ba01435d2b41', 6: '16367aacb67a', 7:
'27042f4e6eca', 8: '80b62dc0bfc3', 9:
'251fec592e5', 10: '596d40a4ed24', 11:
'ba01435d2b41', 12: '3a8f7b6c00a3'}
trigger_check(answers, KEY)
```

On the timer — 8 minutes

November 2022 Paper 2 Q11(a)(b), 6 marks · calculator

The function f is defined by $f(x) = e^{2x}(3x - 4)$, where $x \in \mathbb{R}$.

(a) Find $f'(x)$.

(b) Hence, or otherwise, find the coordinates of the point on the graph of $y = f(x)$ where the tangent is parallel to the line $y = x$. Give your answer to 3 significant figures.

Eight minutes. Part (a) is E3 and should take one; the six marks are mostly in knowing what *parallel to $y = x$* is worth.

```
qt_a = ...          # f'(x)
qt_b = ...          # (x, y), 3 s.f.

verify_derivative('timer (a)', qt_a,
exp(2*x)*(3*x - 4))
verify_where('timer (b)', qt_b, curve(Eq(y,
exp(2*x)*(3*x - 4))), 1, (-2, 3))
```

Solutions

Read these after you have written something down, not before.

Task 1

(a) The tangent's gradient *is* the derivative at the point of contact: $f'(4) = 6$. **A1**

(b) The point of contact lies on the tangent as well as on the curve, so $f(4) = 6 \cdot 4 - 1 = 23$. **A1** This is the whole of the mark: nothing about f is known except through the line.

(c) Chain rule: $h'(x) = f'(g(x))g'(x)$. Now $g(4) = 16 - 12 = 4$, which is the point where we know things about f , and $g'(x) = 2x - 3$ so $g'(4) = 5$:

$$h'(4) = f'(4) \cdot 5 = 6 \cdot 5 = 30.$$

Also $h(4) = f(g(4)) = f(4) = 23$. So

$$y - 23 = 30(x - 4), \quad y = 30x - 97.$$

M1A1A1

The coincidence that makes this question work is $g(4) = 4$: without it $f'(g(4))$ would be a number nobody has. Notice it early and the question is three lines.

Task 2

$$\begin{aligned} g(a) &= (a - r)(a^2 - 2a^2 + a^2 + b^2) \\ &= b^2(a - r), \quad g'(a) \\ &= 0 + a^2 - 2a^2 + a^2 + b^2 \\ &= b^2. \end{aligned}$$

The first factor of g' vanishes at $x = a$, which is why $g'(a)$ is so clean. Then

$$\begin{aligned} y - b^2(a - r) &= b^2(x - a) \\ &\implies y \\ &= b^2x - b^2r \\ &= b^2(x - r). \end{aligned}$$

Setting $y = 0$ and using $b > 0$, so $b^2 \neq 0$, gives $x = r$. **A1A1M1A1M1R1**

The $b > 0$ is not decoration: dividing by b^2 is the step, and it is legal only because the question said so.

Task 3

At $x = -3$: $f(-3) = \frac{9}{2} - 15 + 13 = \frac{5}{2}$, and $f'(x) = x + 5$ so $f'(-3) = 2$. The normal has gradient $-\frac{1}{2}$:

$$\begin{aligned}y - \frac{5}{2} &= -\frac{1}{2}(x + 3) \\&\implies y \\&= -\frac{1}{2}x - \frac{3}{2} + \frac{5}{2} \\&= -\frac{1}{2}x + 1.\end{aligned}$$

A1A1A1M1A1

Five marks for five lines, and four of them are arithmetic. This is what a normal question looks like on Paper 1.

Task 4

$y = x^{-1}$ gives $\frac{dy}{dx} = -x^{-2}$, so at $x = t$ the tangent has gradient $-\frac{1}{t^2}$ and the normal has

$$m_N = \frac{-1}{-1/t^2} = t^2. \quad \mathbf{M1A1}$$

Two marks, and the whole of the rest of that Paper 3 — 24 more marks about how many normals of a given gradient a hyperbola has — rests on this line.

Task 5

$$\begin{aligned}f'(x) &= \frac{2x}{x^2 - 16} \\&= \frac{1}{3} \\&\implies 6x \\&= x^2 - 16 \\&\implies x^2 - 6x - 16 \\&= 0 \\&\implies (x - 8)(x + 2) \\&= 0.\end{aligned}$$

The domain is $x > 4$, so $x = 8$.

M1A1M1A1M1A1

The mark scheme says: *award A0 if the final answer includes additional solutions*. Writing " $x = 8$ or $x = -2$ " is not a partially right answer here; it is a wrong one, because -2 is not in the domain of f at all.

Task 6

$f'(x) = -45e^{-0.5x}$, and setting that to -1 :

$$\begin{aligned}
e^{-0.5x} &= \frac{1}{45} \\
&\implies -\frac{1}{2}x \\
&= -\ln 45 \\
&\implies x \\
&= 2 \ln 45.
\end{aligned}$$

Then

$$y = 90e^{-\ln 45} = \frac{90}{45} = 2.$$

So $Q = (2 \ln 45, 2)$. **A1M1A1A1**

The y -coordinate coming out as a clean 2 is the sign you have not slipped; if it is 1.9998 you have rounded x somewhere you should not have. And $2 \ln 45 = \ln 2025$ is the same answer, which is why the check compares numbers and not strings.

Task 7

Quotient rule, with a common factor pulled out:

$$\begin{aligned}
f'(x) &= \frac{3(2x+a)^2 \cdot 2(x+5)^2 - (2x+a)^3 \cdot 2(x+5)}{(x+5)^4} \\
&= \frac{2(2x+a)^2(x+15-a)}{(x+5)^3}.
\end{aligned}$$

At $x = 1$ that is $\frac{(2+a)^2(16-a)}{108}$, and the angle gives

$$(2+a)^2(16-a) = 108 \tan 70^\circ \approx 296.7.$$

Solving the cubic: $a = 2.73$ and $a = 14.97 \approx 15.0$ (the third root is negative and $a > 0$).

M1A1A1A1

The angle is the whole trick, and it is worth saying out loud: a line at 70° to the horizontal has gradient $\tan 70^\circ \approx 2.75$, not 70 and not $\frac{70}{90}$.

Task 8

$$\begin{aligned} & 8x \\ & + 2y \frac{dy}{dx} \\ & - 24 \\ & + 4 \frac{dy}{dx} = 0 \\ \implies & (2y + 4) \frac{dy}{dx} \\ & = 24 - 8x \\ \implies & \frac{dy}{dx} \\ & = \frac{24 - 8x}{2y + 4} \\ & = \frac{4(3 - x)}{y + 2}. \end{aligned}$$

M1A1A1A1

Two places to lose it: the $4y$ term differentiates to $4 \frac{dy}{dx}$ and not to 4, and the final cancelling is a factor of 2 top and bottom — the printed form has 4 in the numerator, not 8.

Task 9

(a) Write $w = xy$, so $w' = y + x \frac{dy}{dx}$ and $\frac{d}{dx}(w \ln w) = w'(\ln w + 1)$. Differentiating $y = x - xy \ln(xy)$:

$$\frac{dy}{dx} = 1 - \left(y + x \frac{dy}{dx}\right)(1 + \ln(xy)),$$

which rearranges to the printed line.
M1A1A1 Solving for the derivative,

$$\frac{dy}{dx} = \frac{1 - y(1 + \ln(xy))}{1 + x(1 + \ln(xy))}.$$

(b) At $x = 1$ the curve says $y = 1 - y \ln y$, and $y = 1$ satisfies it — it is the only positive solution. Substituting $x = y = 1$: $\ln(xy) = 0$, so

$$\frac{dy}{dx} = \frac{1 - 1 \cdot 1}{1 + 1 \cdot 1} = 0,$$

and the tangent is the horizontal line $y = 1$. **M1A1M1A1A1**

The point is worth dwelling on: the *only* way to get that $y = 1$ is from the curve's own equation. There is no formula for y to substitute into.

Task 10

(a) Differentiating $e^{x+y} = x^2 + y^2$:

$$\begin{aligned}e^{x+y} \left(1 + \frac{dy}{dx} \right) &= 2x + 2y \frac{dy}{dx} \\ \implies \frac{dy}{dx} &= \frac{2x - e^{x+y}}{e^{x+y} - 2y}.\end{aligned}$$

(b) Horizontal means the numerator vanishes: $e^{x+y} = 2x$, so $x + y = \ln(2x)$ and $y = \ln(2x) - x$. Substituting into the curve,

$$\begin{aligned}2x &= x^2 + (\ln(2x) - x)^2 \\ \implies 2x^2 &+ (\ln 2x)^2 \\ &- 2x \ln 2x \\ &- 2x \\ &= 0.\end{aligned}$$

The calculator gives $x = 0.331$ and $x = 1.842$, and then $y = \ln(2x) - x$ gives

$$\begin{aligned}P &= (0.331, -0.743), & Q \\ &= (1.84, -0.538).\end{aligned}$$

(c) Gradient -1 means $2x - e^{x+y} = -(e^{x+y} - 2y)$, that is $x = y$ — the line of

symmetry, which the diagram told you. Then $e^{2x} = 2x^2$ gives $x = -0.451$, and the point is $(-0.451, -0.451)$.

Two things to take away. **The horizontal-tangent condition is about the numerator alone**, and it is one equation in two unknowns until you bring the curve back in. And **there are two points, not one** — the check counts them, and so does the mark scheme.

Task 11

(a) Substitute straight into the relation: with $x = 1$ and $y = \frac{3}{2}$,

$$\begin{aligned} \left(1 + \frac{3}{2}\right) \frac{dy}{dx} &= 1 \\ &+ \frac{3}{2} \\ &- 3 \cdot \frac{9}{4} \\ &= -\frac{17}{4} \\ \implies \frac{dy}{dx} &= -\frac{17}{10} \\ &= -1.7. \end{aligned}$$

(b) Now put $x = 1$, $y = \frac{3}{2}$, $\frac{dy}{dx} = -1.7$ into the second relation:

$$\begin{aligned} \frac{5}{2} \cdot \frac{d^2y}{dx^2} &= 2 \\ &+ \frac{3}{2} \\ &- 1 \cdot (-1.7)^2 \\ &- \left(1 + \frac{21}{2}\right)(-1.7) \\ &= 3.5 - 2.89 + 19.55 \\ &= 20.16, \end{aligned}$$

$$\frac{d^2y}{dx^2} = \frac{20.16}{2.5} = 8.064 = \frac{1008}{125}.$$

M1A1

Where does the second relation come from? Differentiate the first one:

$$\begin{aligned} (2x + y + x \frac{dy}{dx}) \frac{dy}{dx} + (x^2 + xy) \frac{d^2y}{dx^2} \\ = 2x \\ + y \\ + x \frac{dy}{dx} \\ - 6y \frac{dy}{dx}, \end{aligned}$$

and collecting the $\frac{dy}{dx}$ terms on the right gives the printed line, with the $-x(\frac{dy}{dx})^2$

coming from the product rule on $x \frac{dy}{dx}$ inside the bracket.

A footnote on the corpus this practicum was built from: it prints that relation with $+2x\left(\frac{dy}{dx}\right)^2$ instead of $-x\left(\frac{dy}{dx}\right)^2$, which gives the wrong number here. The paper is right; substituting and getting 8.064 is how you tell.

Task 12

$f'(x) = -\sin x$, so $f'(k) = -\sin k$. And $g'(x) = \sec^2 x$, so

$$g'(k) = \sec^2 k = \frac{1}{\cos^2 k} = \frac{1}{\sin k}$$

using the given $\cos^2 k = \sin k$. Therefore

$$f'(k) g'(k) = -\sin k \cdot \frac{1}{\sin k} = -1,$$

so the tangents are perpendicular.

A1A1R1

That substitution is the entire question. Without $\cos^2 k = \sin k$ the product is $-\sin k \sec^2 k$ and stays a function of k ; with it, everything cancels.

For the record, part (c): $\cos^2 k = \sin k$ becomes $1 - \sin^2 k = \sin k$, so $\sin^2 k + \sin k - 1 = 0$ and $\sin k = \frac{-1+\sqrt{5}}{2}$, the positive root.

Task 13

(a) A point (x, y) on $y = mx$ has $m = \frac{y}{x}$, and since the gradient of the line is m , the gradient of L is $\frac{y}{x}$. **A1**

One mark, and it is the hinge of the whole 31-mark investigation: written in x and y , the gradient of the family can be fed straight into perpendicularity, and $\frac{dy}{dx} = -\frac{x}{y}$ solves to $x^2 + y^2 = k$.

(b) Differentiating each family implicitly,

$$\begin{aligned} y^2 &= 4a^2 - 4ax \\ \implies 2y \frac{dy}{dx} &= -4a \\ \implies \frac{dy}{dx} &= -\frac{2a}{y}, \end{aligned}$$

$$y^2 = 4b^2 + 4bx \implies \frac{dy}{dx} = \frac{2b}{y}.$$

At M, where $y = 2\sqrt{ab}$, these are $-\frac{a}{\sqrt{ab}}$ and $\frac{b}{\sqrt{ab}}$, and

$$-\frac{a}{\sqrt{ab}} \cdot \frac{b}{\sqrt{ab}} = -\frac{ab}{ab} = -1.$$

M1A1M1A1A1

Both parabolas are drawn as $y^2 = \dots$, so implicit differentiation is not a choice made for elegance — solving for y would put a square root in everything and a $\pm$ in front of it.

Task 14

(a) $f'(x) = -2(x - h)$ and $g'(x) = e^{x-2}$, so $g'(3) = e$. A common tangent at $x = 3$ means the gradients agree there:

$$\begin{aligned} -2(3 - h) &= e \\ \implies 3 - h &= -\frac{e}{2} \\ \implies h &= 3 + \frac{e}{2} \\ &= \frac{e + 6}{2}. \end{aligned}$$

A1M1A1

(b) And the values agree there too — that is the second half of *common tangent*:

$$-(3-h)^2 + 2k = e + k.$$

With $3-h = -\frac{e}{2}$ the left side is $-\frac{e^2}{4} + 2k$, so

$$k = e + \frac{e^2}{4} \approx 4.57.$$

M1A1A1

Everything about this question is in the phrase *common tangent*: two conditions, two unknowns, one at a time.

Task 15

Let $y = \log_a x = \frac{\ln x}{\ln a}$. Tangency to $y = x$ needs the gradient to be 1:

$$\frac{dy}{dx} = \frac{1}{x \ln a} = 1 \implies x = \frac{1}{\ln a},$$

and needs the point to be on both lines, so $\log_a x = x$:

$$\frac{\ln x}{\ln a} = x.$$

Substituting $x = \frac{1}{\ln a}$ into that gives $\ln\left(\frac{1}{\ln a}\right) = 1$, hence $\frac{1}{\ln a} = e$ and so

$$x = e,$$

$$P = (e, e),$$

$$\ln a = \frac{1}{e},$$

$$a = e^{1/e} \approx 1.4447.$$

A1M1A1M1A1A1M1A1

The $1.4 \leq a \leq 1.5$ in the question is a check, not a method: it tells you which of the plausible answers you are heading for. And $P = (e, e)$ is on $y = x$ by construction, so finding x finds both coordinates.

On the timer

(a) Product rule:

$$f'(x) = 2e^{2x}(3x - 4) + 3e^{2x} = e^{2x}(6x - 5).$$

M1A1A1

(b) *Parallel to $y = x$* means gradient 1:

$$e^{2x}(6x - 5) = 1.$$

Nothing algebraic will solve that; the calculator gives $x = 0.863$, and

$$y = e^{2(0.863)}(6(0.863) - 4) = -7.93.$$

So the point is $(0.863, -7.93)$. **M1A1A1**

Substitute into f , not into f' . It is the single most common way to lose the last mark on a question of this shape, and it is entirely mechanical to avoid.

Key to the recognition drill

1 **implicit** — $y^2 = x^3 + x$ cannot be split, and the answer keeps both letters. 2 **condition** — the tangent is not asked for; *parallel* is an equation on k . 3 **point** — the gradient is given as 1, and the point is the answer. 4 **normal** — the word is in the question, and $-1/m$ is the whole of it. 5 **tangent** — a gradient at a named point of a written function. 6 **second** — differentiate the relation again, then substitute it into itself. 7 **right** — two curves, one point, product -1 . 8 **itangent**

— a tangent, but to a curve given by an equation. 9 point — gradient $\frac{1}{3}$ given, x -coordinate wanted. 10 condition — *show that the line is tangent*: touching is the claim. 11 tangent — a tangent at a named point, with a chain rule inside. 12 implicit — take logs of both sides and differentiate; x is in the base and the exponent at once.

Four of these are worth arguing about, and the argument is the point.

1 could be itangent. It is not: nothing is asked about a line. The question stops at $\frac{dy}{dx}$, and that is rung 4. Part (f) of the same paper *is* itangent, and that is question 8 here.

3 and 9 look different and are the same. One says *parallel to* $y = x$ and one says *gradient* $\frac{1}{3}$; both are a number, and both want a point back. The phrasing is the only thing that changes.

10 could be tangent. The word is there, and the technique is not: nothing is being found. What is being *established* is that

touching happens, which needs two conditions or a double root — rung 8.

12 could be point, because a maximum is where the gradient is zero. But the gradient of $(S/x)^x$ does not exist until you take logarithms, and the taking of logarithms is the question. The zero is the last line, not the first.

Where the marks went, across the topic

technique	marks	share
Touching as a condition	23	18%
The curve is an equation	21	16%
Tangent to such a curve	20	15%
Gradient given, find the point	17	13%
Differentiate the relation again	15	11%
Tangent at a known point	14	11%
Right angle as a condition	14	11%
Normal	7	5%

Three things stand out.

The two ends of the ladder are worth more than the middle. Rungs 1–3, the

ones every course drills, carry 38 marks. Rungs 7–8, where the line stops being the answer, carry 37 — and almost nobody practises them, because they do not look like tangent questions at all until you notice the word *common*.

Rungs 4–6 are the biggest single block at 56 marks, and the calculus in them is one rule: y is a function of x , so it brings a $\frac{dy}{dx}$ with it. Everything else is bookkeeping — and the bookkeeping is where the marks go.

Forty-two of the 131 marks are Paper 3. The same shape E3 had: this material does not appear as a five-mark question at the end of a paper, it appears as one step of a twenty-eight mark investigation, over and over. The November 2023 investigation on orthogonal families and the November 2025 one on normals to hyperbolas are 57 marks between them, of which this topic claims 8 — the other 49 belong to E7, B4 and A5, and that is the honest picture of how Paper 3 is built.